

Aung Khant Ko

Bethlehem, Pennsylvania | +1 484-750-7861 | aungkko.edu@gmail.com

LinkedIn: linkedin.com/in/aungkko | GitHub: github.com/akkexd

EDUCATION

Thomas Edison State University

Bachelor of Arts in Computer Science

June 2026

Trenton, New Jersey

King Mongkut's Institute of Technology Ladkrabang

Visiting Student, Robotics and Artificial Intelligence Engineering

Jul 2023 – Sep 2024

Bangkok, Thailand

EXPERIENCE

Visiting Research Collaborator

Lehigh University, CSE (Advisor: **Prof. Corey Montella**)

Jun 2026 – Present

Bethlehem, PA

- Integrated Mech 0.3.5 reactive dataflow into mecanum robot control.
- Built Mech from source on aarch64 Raspberry Pi 5.
- Benchmarked Mech versus Python and C++ kinematics costs.
- Designed five-experiment latency and jitter measurement suite.

Robotics Software Developer

Independent Robotics Research Project, Thomas Edison State University

Nov 2025 – May 2026

Bethlehem, PA

- Developed a ROS2 Jazzy autonomous navigation stack for a mecanum robot.
- Implemented EKF fusion to improve state estimation under wheel slip on high-friction carpet.
- Configured Nav2 with AMCL, NavFn, and DWB for localization, planning, and navigation.
- Improved carpet navigation with goal management and deadzone compensation.
- Integrated LiDAR and depth sensors for mapping and perception.

Robotics (AI) Research and Development Intern

DHA Siamwalla Ltd.

Apr 2024 – Sep 2024

Bangkok, Thailand

- Researched autonomous mobile robots, sensor integration, and SLAM for warehouse automation.
- Built a TurtleBot3 ROS1 prototype for autonomous warehouse navigation and task execution.
- Implemented A*-based planning and costmaps for obstacle-aware navigation.
- Integrated TF, odometry, localization, and sensor pipelines for end-to-end robot autonomy.
- Developed an OpenCV stereo vision pipeline for obstacle detection and localization.
- Tested the system in warehouse aisle navigation and real-time obstacle avoidance scenarios.

PUBLICATIONS

- [1] Ko, A. K. (2026). Practical navigation of a mecanum-wheel robot on high-friction carpet: Motor deadzone compensation, rotation asymmetry, and multi-phase goal management. *engrXiv* (Preprint). <https://doi.org/10.31224/7347>

PROJECTS

- **Reactive Dataflow Control for Mecanum Robot (Mech 0.3.5 Case Study)** — Mecanum-wheel robot control via reactive dataflow oracle benchmarked against Python/C++.
- **Autonomous Mobile Robot (ROS2 Mecanum)** — ROS2 Jazzy mecanum robot on Raspberry Pi 5 with EKF fusion, Nav2, and LiDAR/depth sensing.
- **Sensory Floor Mat for Pediatric Rehabilitation** — Designed a multisensory therapy platform with textured tiles and movement-based games at Good Shepherd Rehabilitation.
- **TurtleBot3 Stereo-Vision Obstacle Avoidance** — ROS1 TurtleBot3 with stereo vision, localization, and path planning for warehouse navigation.
- **Robot End-Effector Design with ABB IRB1200** — Custom pen end-effector for precise line-

drawing tasks on an industrial robot arm.

- **A* Path Planning for TurtleBot3** — A*-based indoor path planning with obstacle avoidance and RViz visualization.
- **Line Following Robot (STM32)** — Embedded STM32 robot with sensor-based line tracking and real-time control.

CERTIFICATIONS

Robotics Software Engineer Nanodegree

Udacity, Aug 2025

Modern Robotics Specialization (6 courses)

Northwestern University / Coursera, Jun 2025

Deep Learning Specialization (5 courses)

DeepLearning.AI, 2020–2024

TECHNICAL SKILLS

Languages: Python, C++, C, MATLAB

Robotics: ROS2, ROS, Nav2, SLAM, Cartographer, Gazebo, Sensor Fusion, EKF, Mecanum Kinematics

AI / ML: Deep Learning, CNNs, TensorFlow, Computer Vision, Object Detection, NLP

Tools & Platforms: Linux (Ubuntu), Git, Arduino, Intel RealSense, RPLIDAR, STM32

Design: PCB Design (EasyEDA), 3D Modeling (Autodesk Inventor, Fusion 360, Tinkercad), Circuit Analysis (Logisim)